



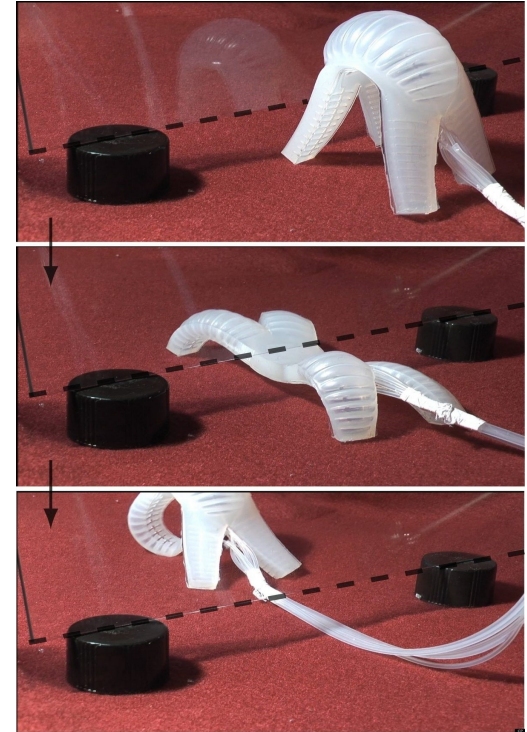
Multi-material soft robotics based on granular and layer jamming

Sophia Lin and Therese Joseph

Supervisor: David Howard

Soft robotics

- Why use soft materials?
 - flexible and compliant = safer
 - robust to impact and collision
 - don't require explicit control
- Applications to medicine and manufacturing (can grab and manipulate delicate and irregular shapes)
- Suitable for unstructured environments (i.e. exploring terrains) (can absorb shock, store energy in springy joints)
- Biomimicry
- Machine learning to determine shapes



<https://www.dailysabah.com/technology/2018/12/04/facebook-has-been-working-on-soft-robotics-since-last-year-looking-for-hires-report>

What is jamming?

Compliant, fluid-like

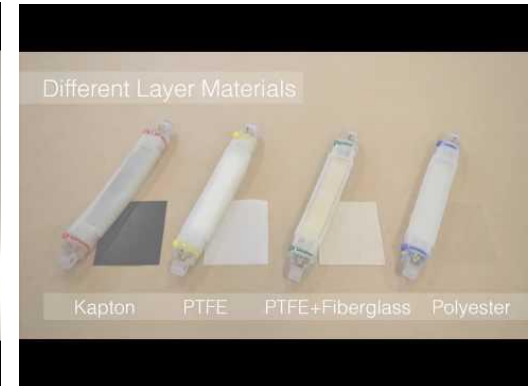


Solid



Universal Grippers with granular jamming

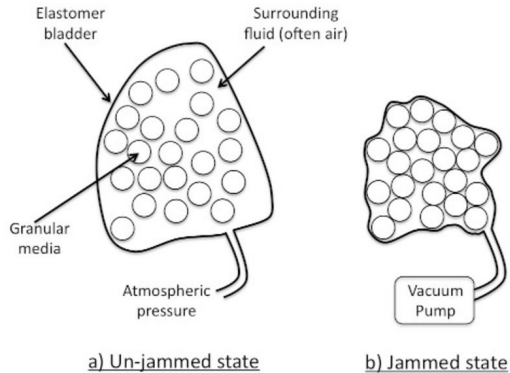
https://youtu.be/ZKOl_IVDPpw



Variable stiffness linear brake with layer jamming

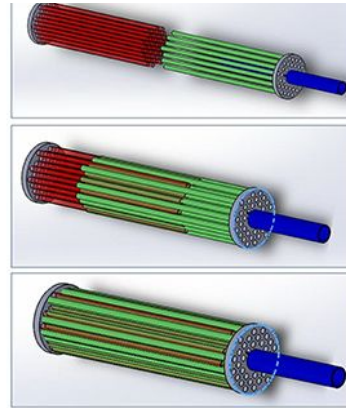
<https://youtu.be/lvoDzVHRzU0>

Types of jamming actuators



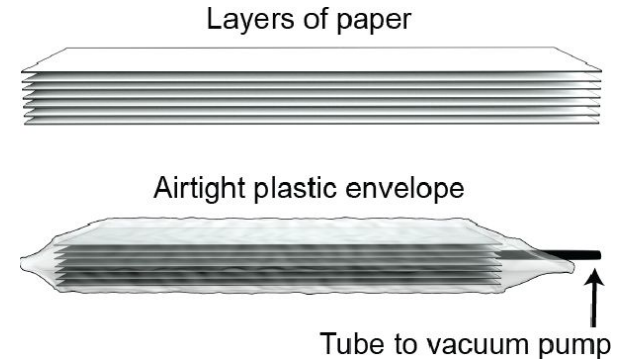
Granular jamming

<https://www.researchgate.net/figure/A-diagram-illustrating-the-basic-concept-behind-using-granular-jamming-for-designed-fig81-279815778>



Fibre jamming

<https://www.semanticscholar.org/paper/Achieving-Mechanical-Versatility-in-Robots-and-Narang/e15ae0f35f577d8ca39dad750272e96d7f511b4e>



Layer jamming

<https://www.frontiersin.org/articles/10.3389/frobt.2019.00012/full>

The Connex3 Objet500 Printer

- Digitally blends photopolymers at different ratios
- Able to vary shore stiffnesses, flexibility, translucency and heat resistance
- Achieves a range of textures such as rubber and traditional hard plastic
- Layer resolution of 16 microns



<https://www.stratasys.com/3d-printers/objet-350-500-connex3>

Our projects

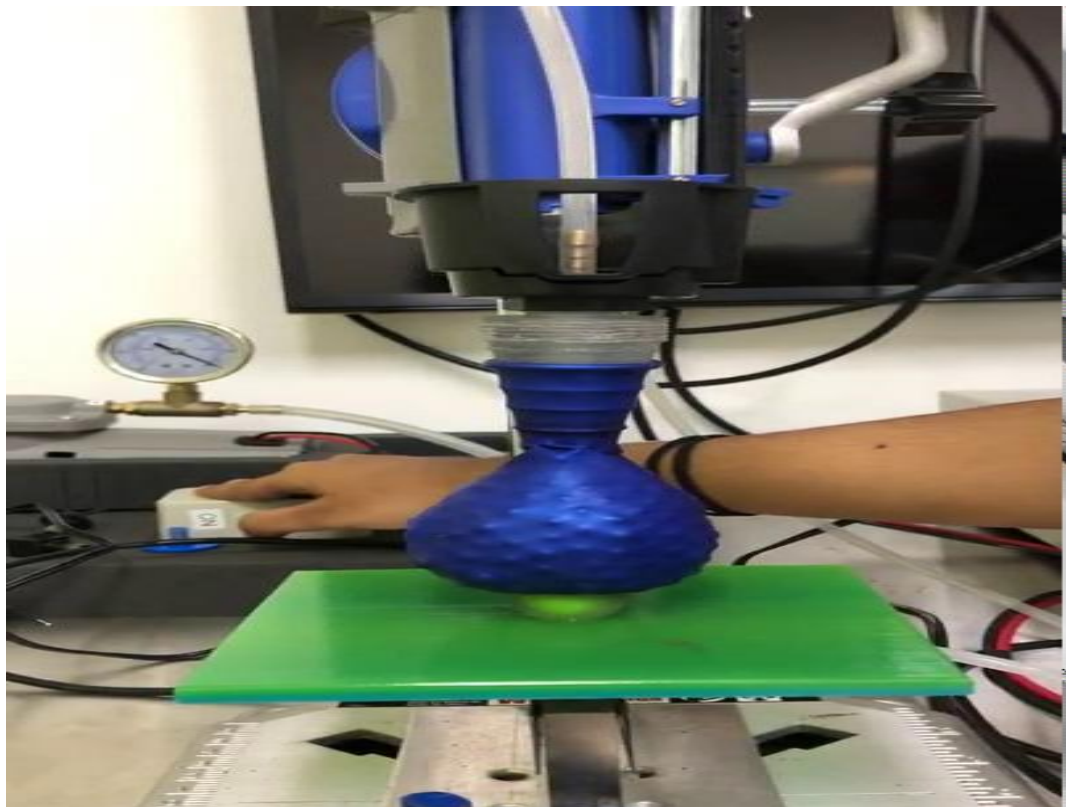
1. Grain characterisation
2. Evolving jamming grippers using a genetic algorithm
3. Exploration of layer jamming

1. Grain characterisation

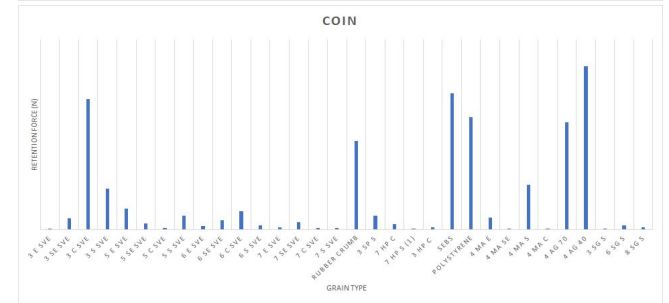
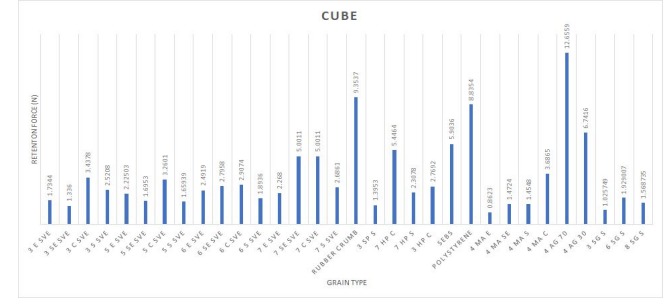
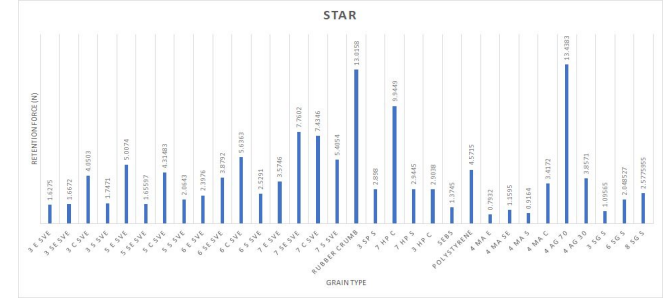
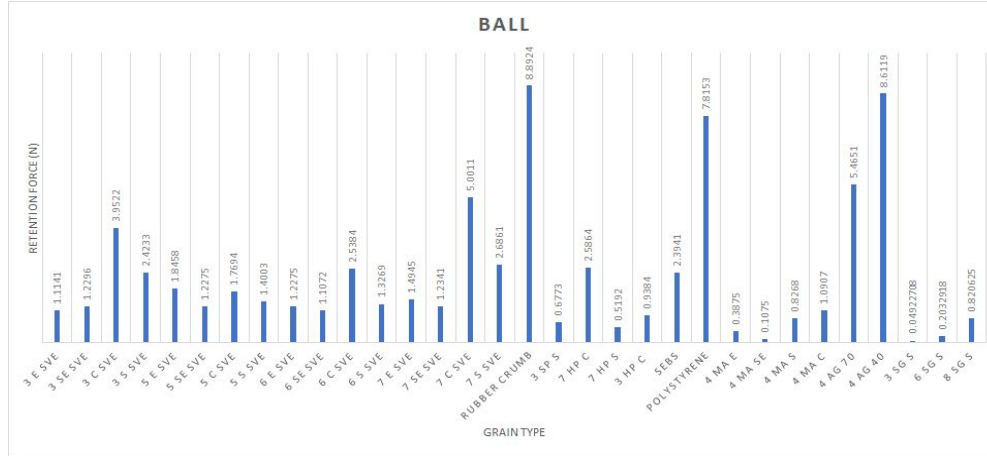
Purpose:

To create a data set characterising the pull-off force of grippers with various grain types





Average retention force for each grain type and test object



2. Evolving jamming grippers using genetic algorithm

Continuing the work of Daniel Furby and James Brett

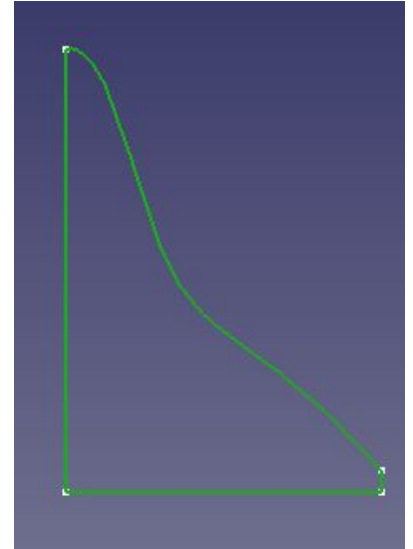
Purpose:

To optimise the geometry of a gripper to pick up a test object

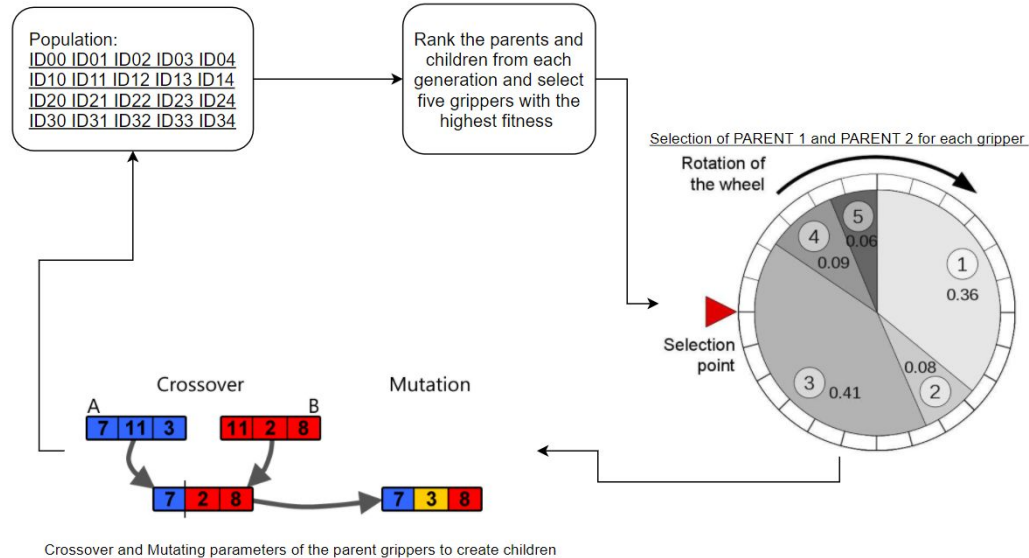


Creating the grippers

- There are 3 parameters
 - Height (30-60mm)
 - Radius (25-40mm)
 - 2-6 control points
- The control points are used to create a b-spline, which is revolved 360 degrees around the origin to create the membrane
- The membrane and adaptor base is created as an stl file for printing

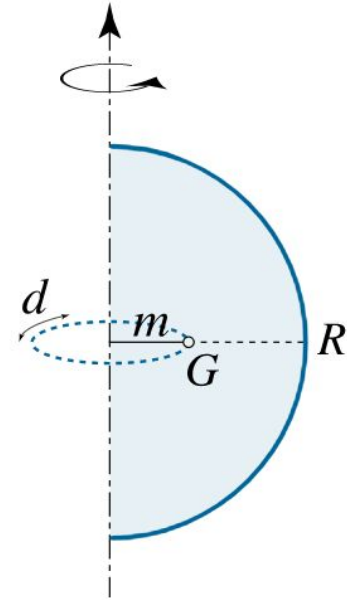


How the genetic algorithm works



The two types of evolution

- We completed two sets of evolution in parallel
 - 1. Fixed volume of the grippers at $100,000 \text{ mm}^3$
 - 2. Variable volume
- To calculate the volume of each gripper, we used Pappus' Centroid Theorem:
 - We calculate the centroid of the cross-section and the circumference of its path, and multiply it by the area of the cross-section to find the volume
- To get grippers with equal volume, we generate a gripper, and change the height and radius by a particular percentage.



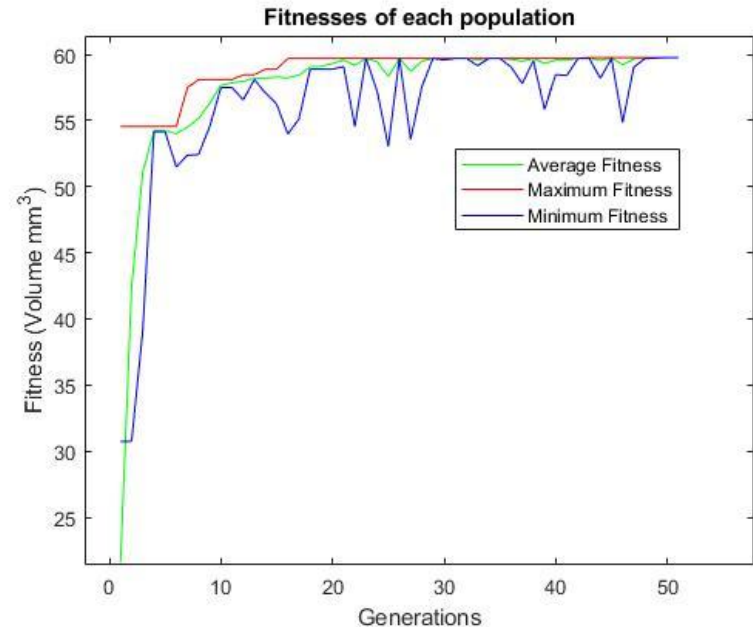
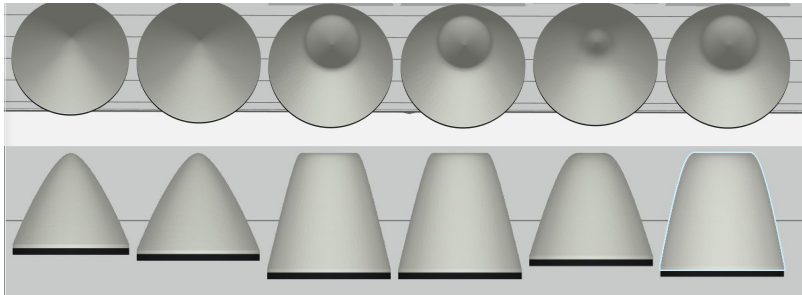
[Pappus's Theorem \(math24.net\)](http://math24.net)

Testing the algorithm

Setting fitness = height

(Height is measured from the base to the centre of the gripper)

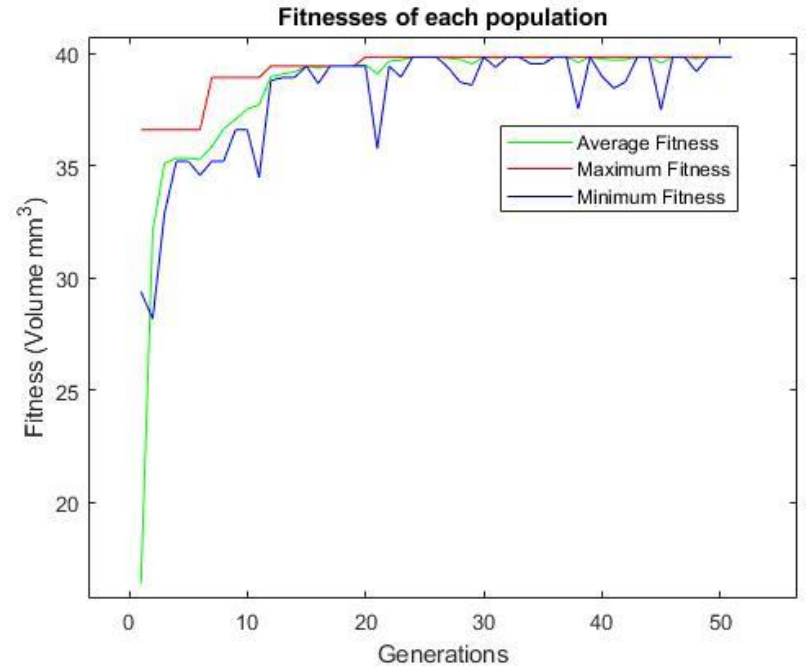
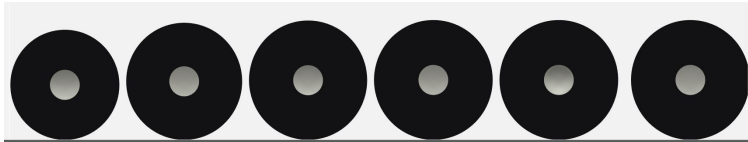
The figure below shows the fittest individuals from every 10 generations.



Testing the algorithm

Setting fitness = radius

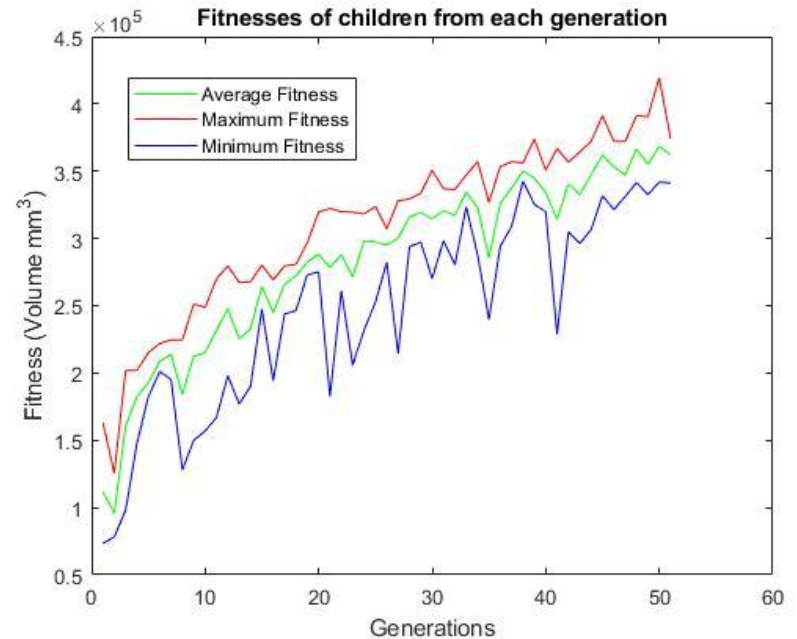
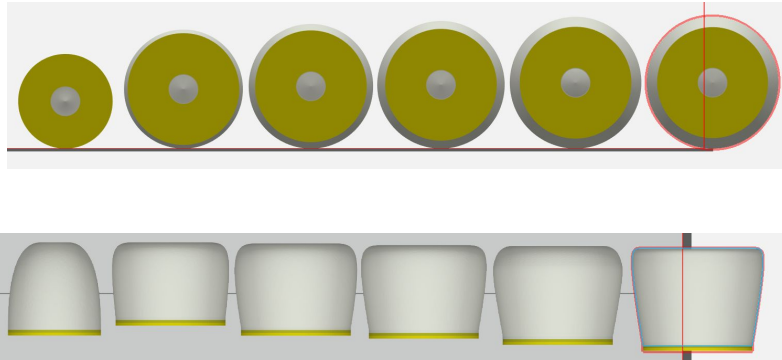
The figure below shows the fittest individuals from each 10 generations.



Testing the algorithm

Setting fitness = volume

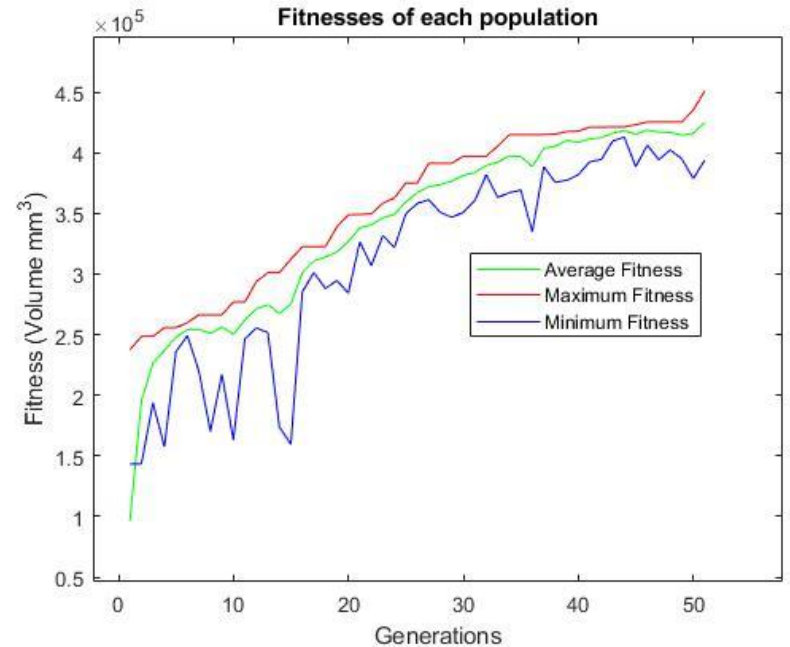
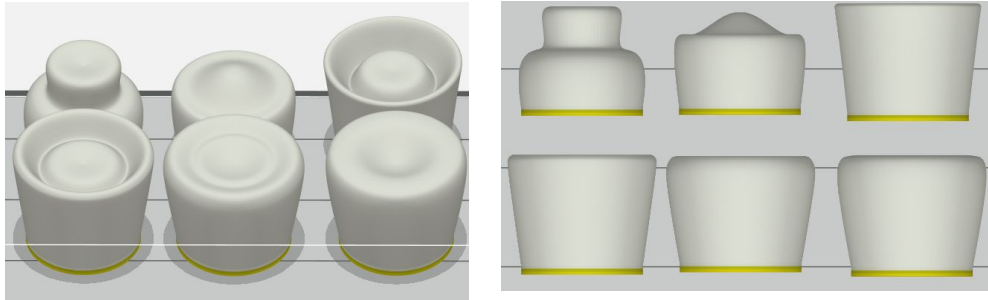
The figure below shows the fittest individuals from each 10 generations.



Testing the algorithm

Setting fitness = volume with fixed height and radius

The figure below shows the fittest individuals from each 10 generations.

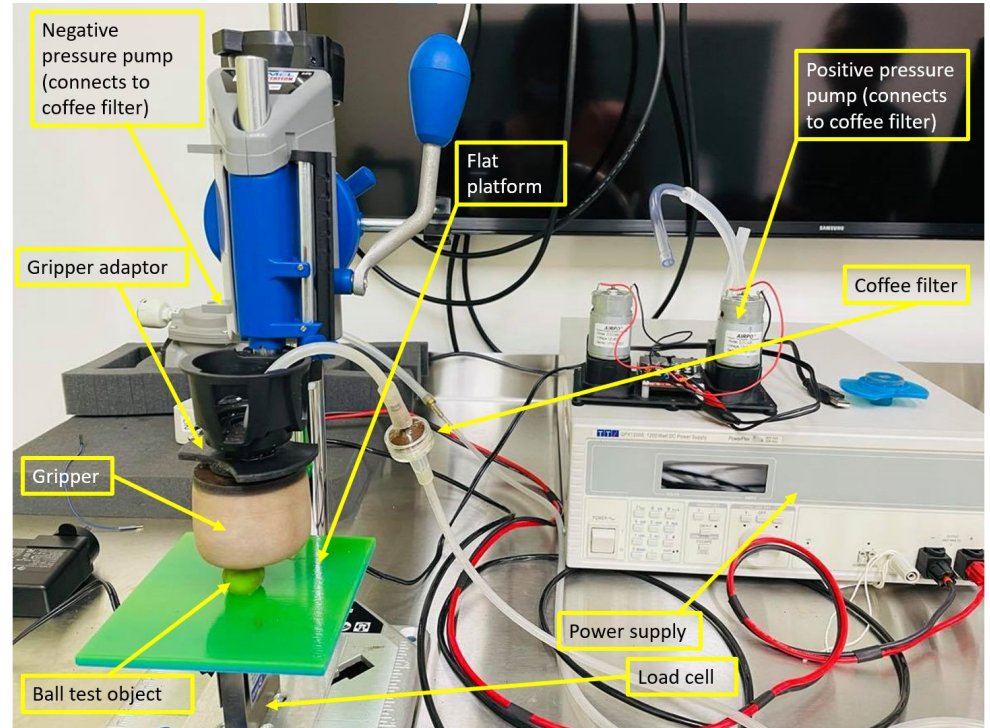


Our testing method

Printing -> Cleaning -> Testing -> Data Entry

Troubleshooting the testing method:

- Pump pressure
- Changing texture due to support type
- Gripper adaptor





Challenges we faced

- FreeCAD volume calculator
 - Use Pappus' Centroid Theorem to calculate volume
- Inconsistent pressure
 - Vacuum regulator
- Surface texture
 - Only use one kind of support
 - No water bath
 - Stickiness
- Delicate, tears
 - Cleaning method - waterjet the outside, pliers for the inside
- Slow code
 - Got rid of the unnecessary parameters that we weren't testing with
 - Equal volume - changed the jumps closer to the volume target



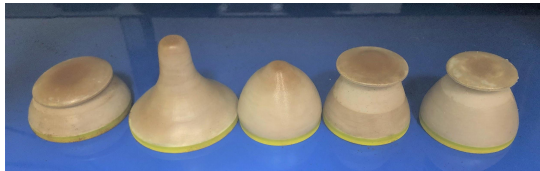
***** Video of testing**

Equal volume jamming grippers

Generation 0



Generation 1



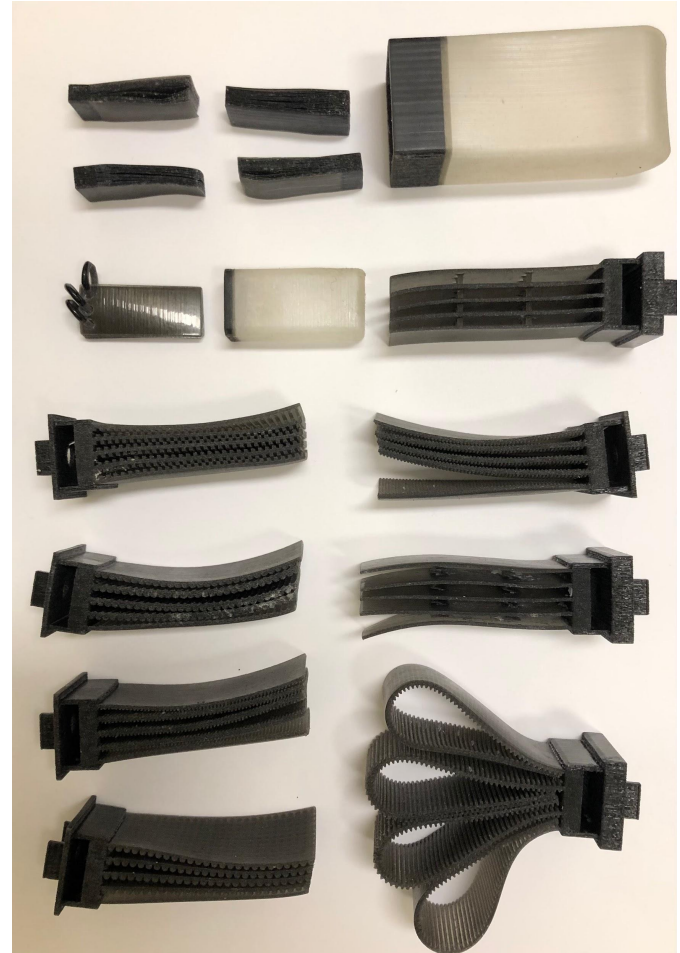
Variable volume jamming grippers

Generation 0



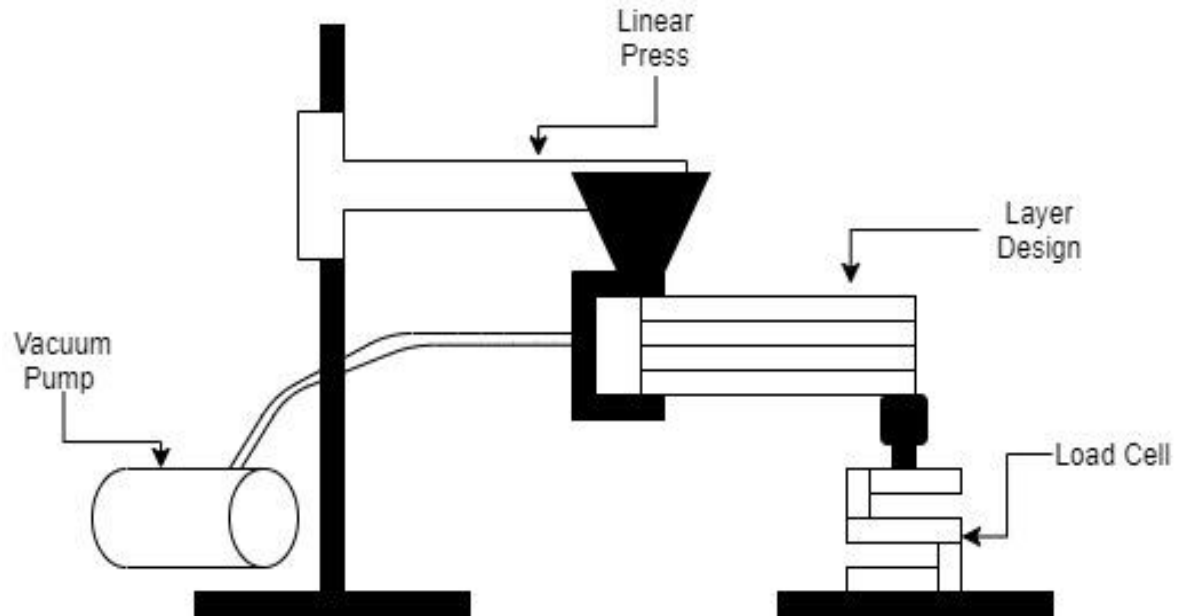
3. Exploration of layer jamming

Purpose:
To come up with a novel design for layer jamming



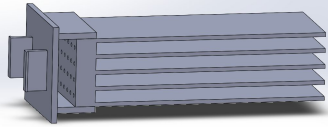
Stiffness testing

We measure stiffness of the layers by deflecting the layers by a fixed amount and measuring the applied force to the load cell.

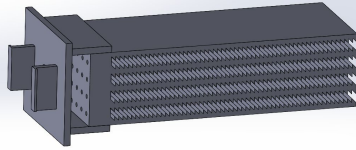




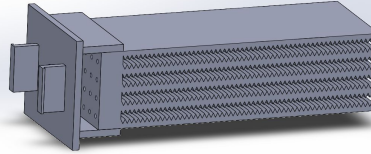
Interlocking designs for increased stiffness



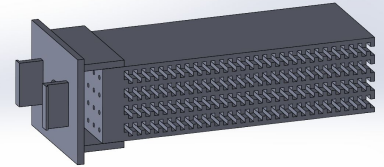
Flat Layers



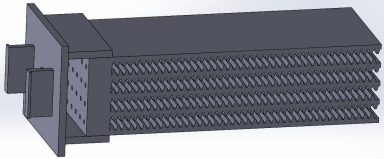
Right angle triangles



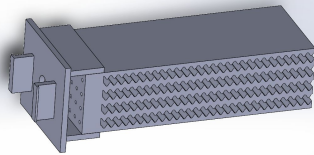
Equilateral triangles



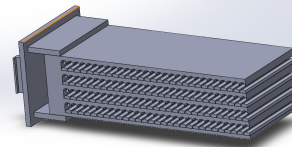
Rectangles



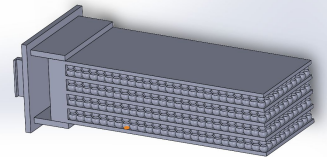
Trapeziums



Sine Wave



Semi-Circle

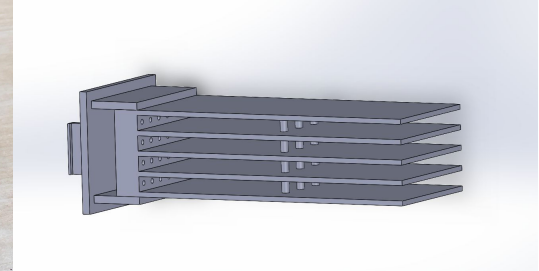
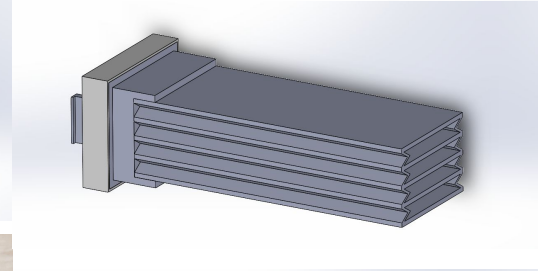
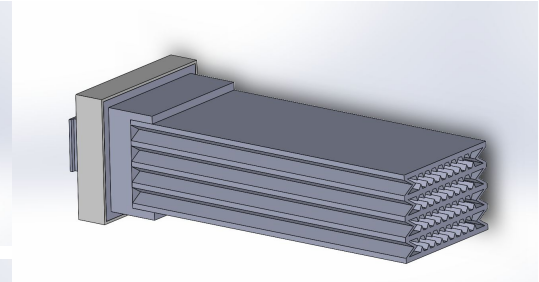
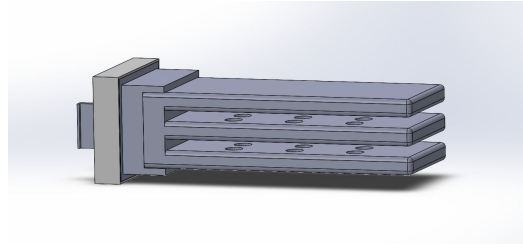
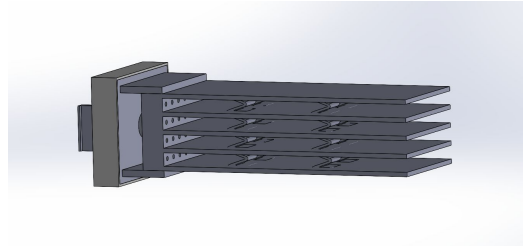


Molecular Velcro

Mechanical unjam and positive pressure

The aim is to maximise the change in stiffness when the vacuum is applied.

We came up with designs which were able to automatically unjam once the vacuum was removed.



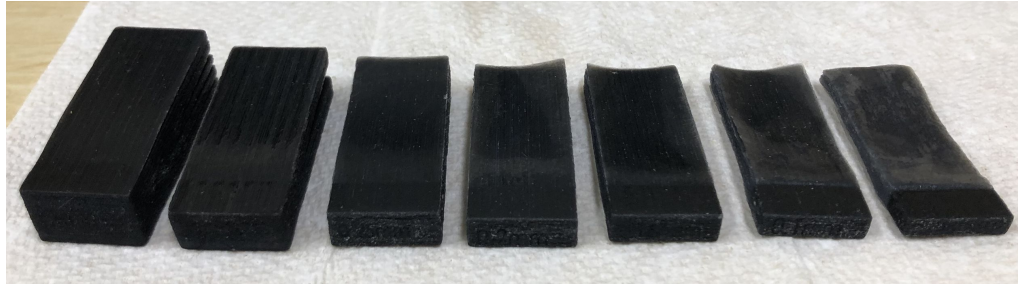


Challenges we faced

- Sticky texture of agilus
 - Silica gel
 - Talcum powder
 - No water bath
- Testing method without resources
-

Micron layers

- We investigated the printer's capabilities in printing thin layers
- Various layer thicknesses were printed in 85, 95 and 100 shore hardness material from 500-50 micron thickness
- Stiffness tests were performed when unjammed (without vacuum) and jammed (with vacuum)



*Layer thickness from left to right: 0.5mm, 0.333mm, 0.25mm, 0.2mm, 0.167mm, 0.1mm, 0.05mm
Each have 15 layers with 0.2mm separation*





Applications of our designs

- Shock absorbency around joints
- Layer Grippers
- Variable stiffness ankle joint for robots

Repeatable mechanical deformation



Thank you!